PROJECT DEVELOPMENT PHASE

FORMULA 1 RACE INTELLIGENCE DASHBOARD

Tableau Dashboard • Tableau Story • Web Integration

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Project documentation with actual dashboard screenshots and cropped visualizations

1. Project Overview

Formula 1 Race Intelligence Dashboard is an interactive data-visualization project designed to explore race records, driver performance, team performance, tyre compound usage, podium achievements, reliability and race-hosting locations. The dashboard was developed in Tableau and published on Tableau Public. A companion web interface provides access to the dashboard and story.

Project links

- **Tableau Public Dashboard:** https://public.tableau.com/app/profile/gouri.sharma8417/viz/f1tableauproject/F1RACEINTELLIGENCEDASHBOARD
- **Tableau Story:** https://public.tableau.com/views/f1tableauproject/Formula1RaceIntelligence
- **GitHub Repository:** https://github.com/gourisanjaysharma-source/F1-Race-Intelligence
- **Live Website:** https://f1-race-intelligence-mhgm.onrender.com/

2. Dashboard Snapshot

The screenshot below is cropped from the supplied Tableau screenshot. The Windows taskbar is excluded.

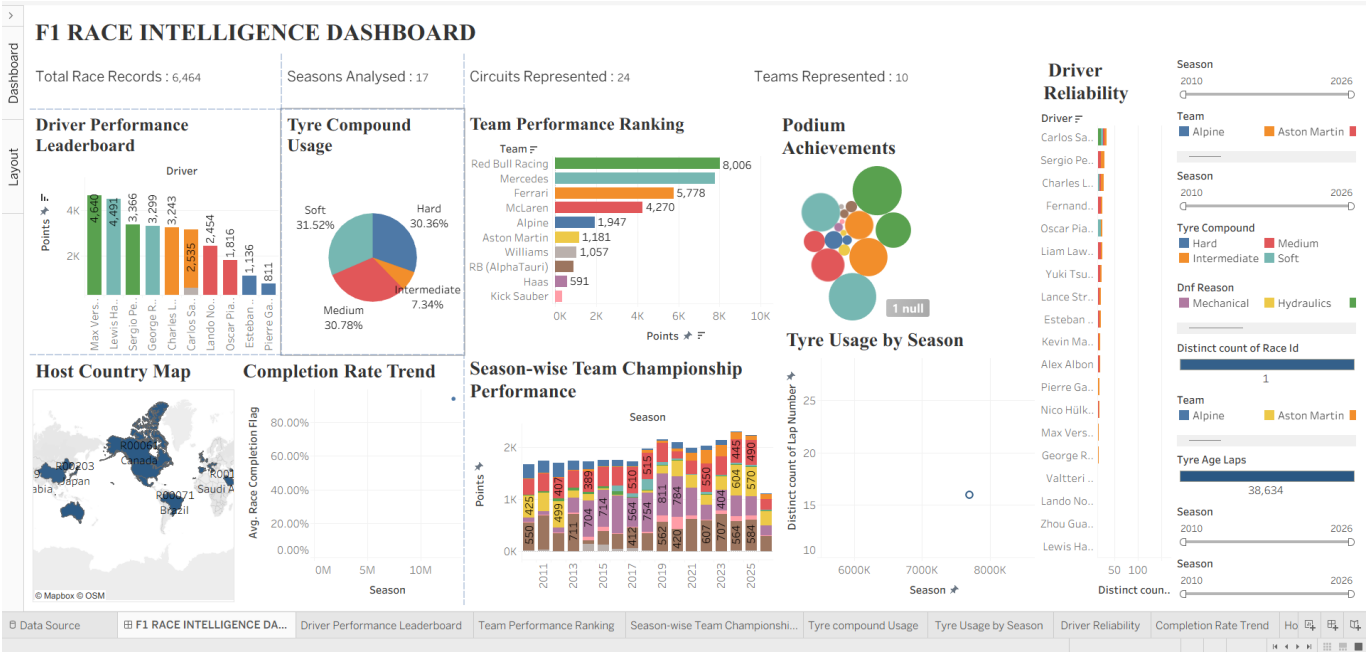


Figure 1. Formula 1 Race Intelligence Dashboard — cropped screenshot, without Windows taskbar.

The displayed dashboard reports 6,464 race records, 17 seasons analysed and 24 circuits represented. These are the values visible in the provided screenshot.

3. Business Questions and Visualisation Report

The following questions are answered or explored by the visualizations visible in the actual dashboard. Descriptions are limited to what can reasonably be read from the screenshot; charts with unclear aggregation are flagged rather than over-interpreted.

1. Driver Performance Leaderboard

Business question: Which drivers have accumulated the most points in the dataset?

Visualization: Vertical bar chart comparing points by driver.

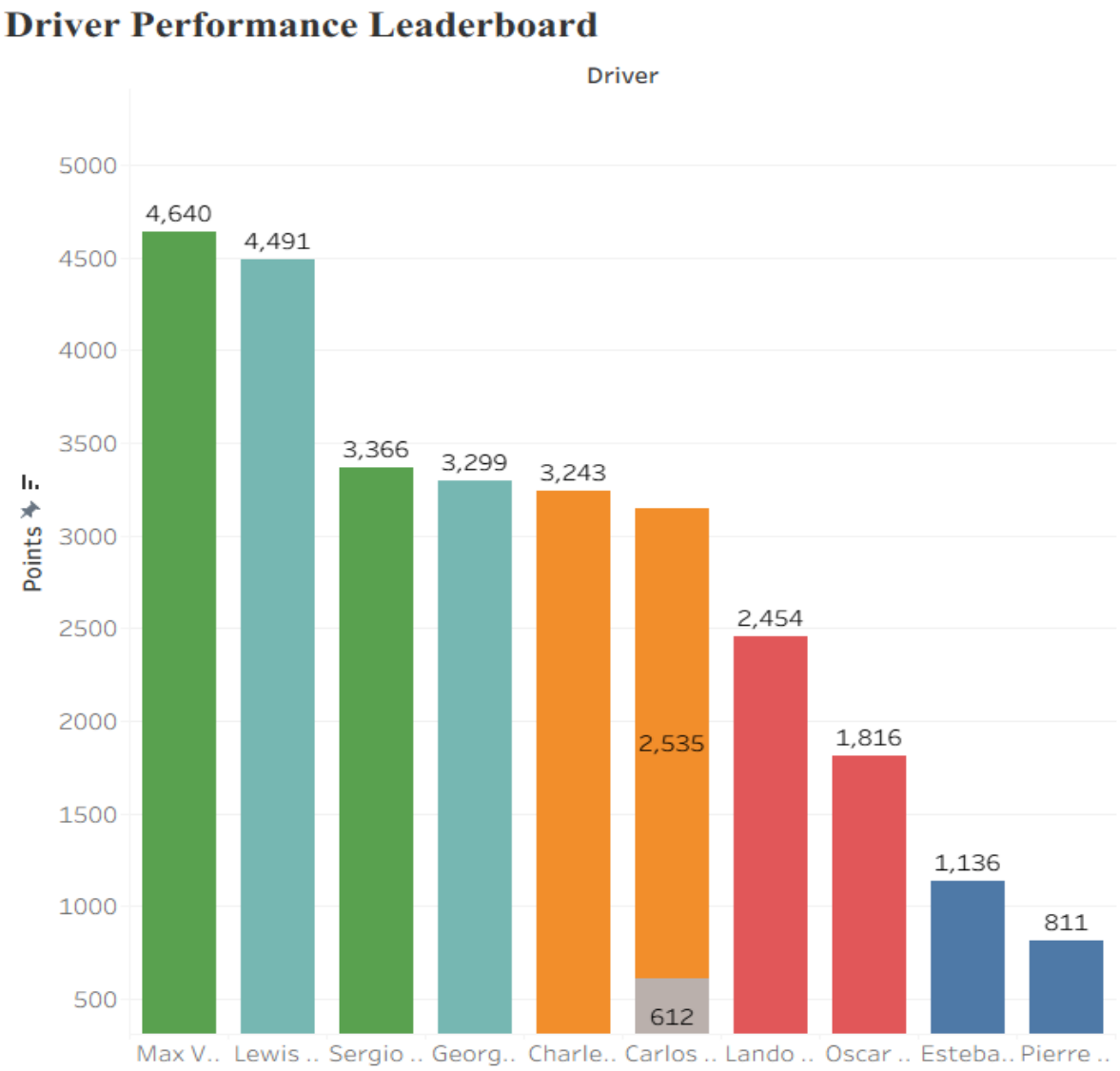


Figure 2. Driver Performance Leaderboard — cropped from the supplied dashboard screenshot; taskbar excluded.

What it shows: The leaderboard places Max Verstappen, Lewis Hamilton and Sergio Perez among the leading drivers in the displayed data.

Interpretation note: The screenshot does not clearly specify whether the points are aggregated across all selected seasons or a filtered subset; interpret rankings within the dashboard's current filter state.

2. Tyre Compound Usage

Business question: How is tyre-compound usage distributed across the recorded data?

Visualization: Pie chart showing the proportion of Soft, Medium, Hard and Intermediate compounds.

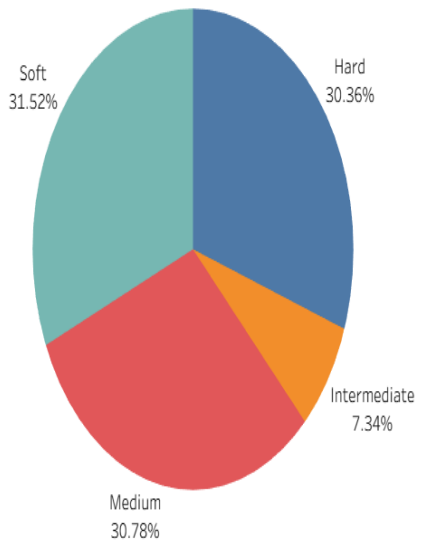


Figure 3. Tyre Compound Usage — cropped from the supplied dashboard screenshot; taskbar excluded.

What it shows: The displayed proportions are Soft 31.52%, Medium 30.78%, Hard 30.36% and Intermediate 7.34%. Soft, Medium and Hard usage are relatively close in this view.

3. Team Performance Ranking

Business question: Which teams have the highest points in the displayed dataset?

Visualization: Horizontal bar chart ranking teams by points.

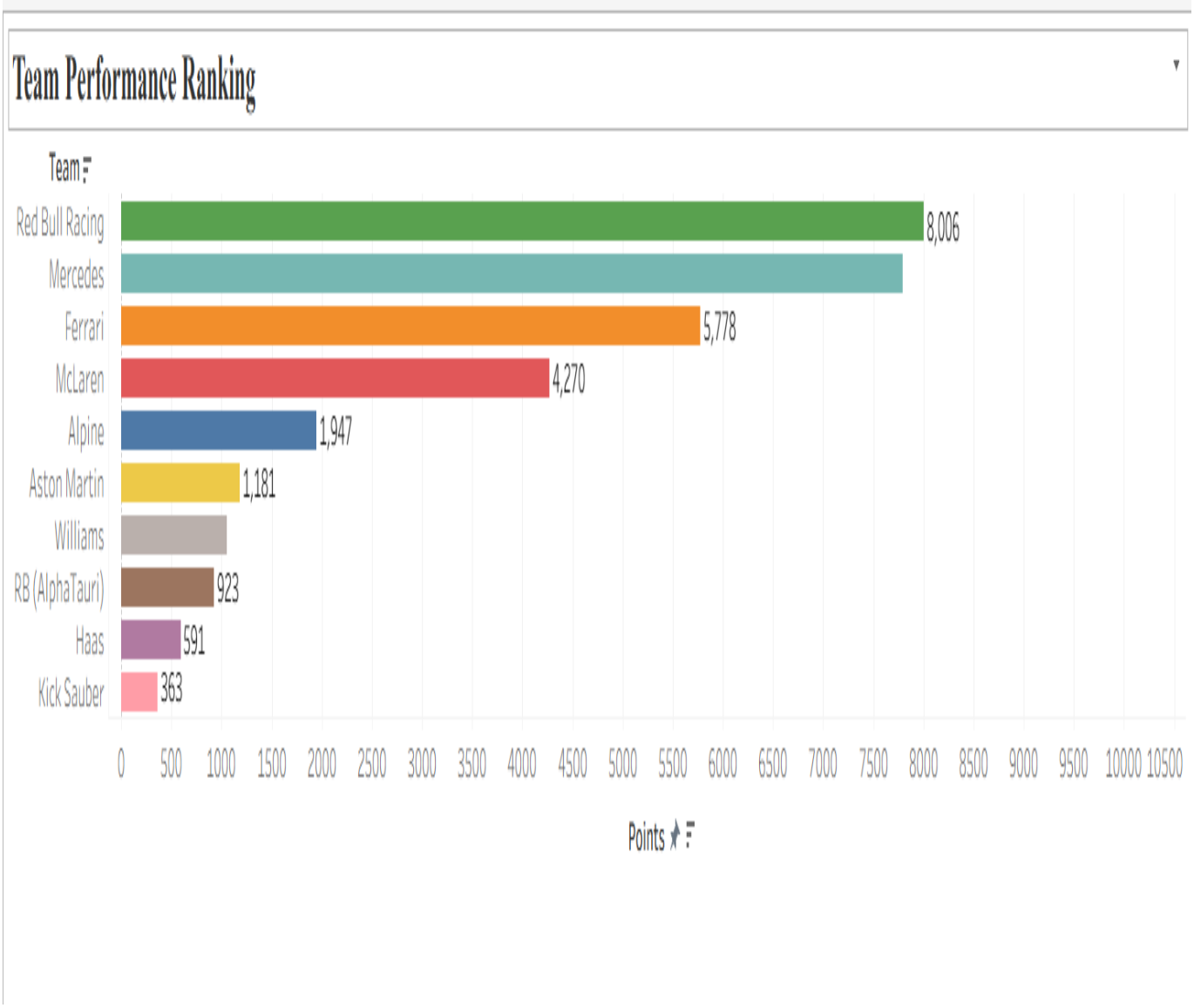


Figure 4. Team Performance Ranking — cropped from the supplied dashboard screenshot; taskbar excluded.

What it shows: Red Bull Racing is the highest bar in the screenshot, followed by Mercedes and Ferrari. The ranking provides a quick team-level comparison for the active filters.

4. Podium Achievements

Business question: How do podium achievements compare across drivers or categories?

Visualization: Bubble chart using bubble size to distinguish podium-achievement values.

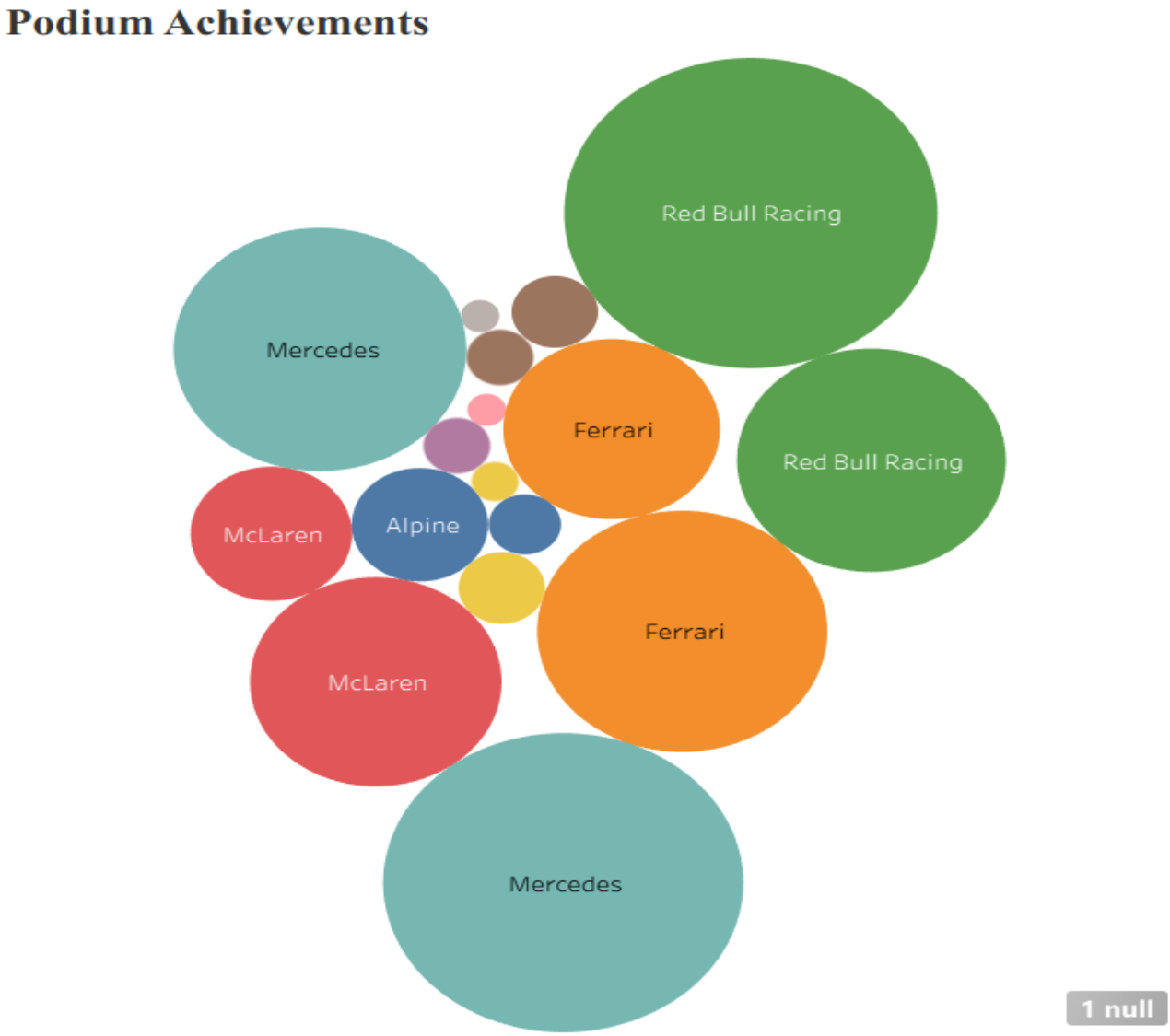


Figure 5. Podium Achievements — cropped from the supplied dashboard screenshot; taskbar excluded.

What it shows: The visualization emphasizes differences in the displayed podium-achievement measure through bubble size. The screenshot includes a '1 null' indicator, so null data should be reviewed before drawing precise conclusions.

Interpretation note: The bubble labels are not fully legible in the supplied screenshot; avoid reporting exact values from this crop.

5. Driver Reliability and DNF Reasons

Business question: What reliability-related retirement reasons appear in the dataset?

Visualization: Driver reliability view alongside a DNF Reason filter.

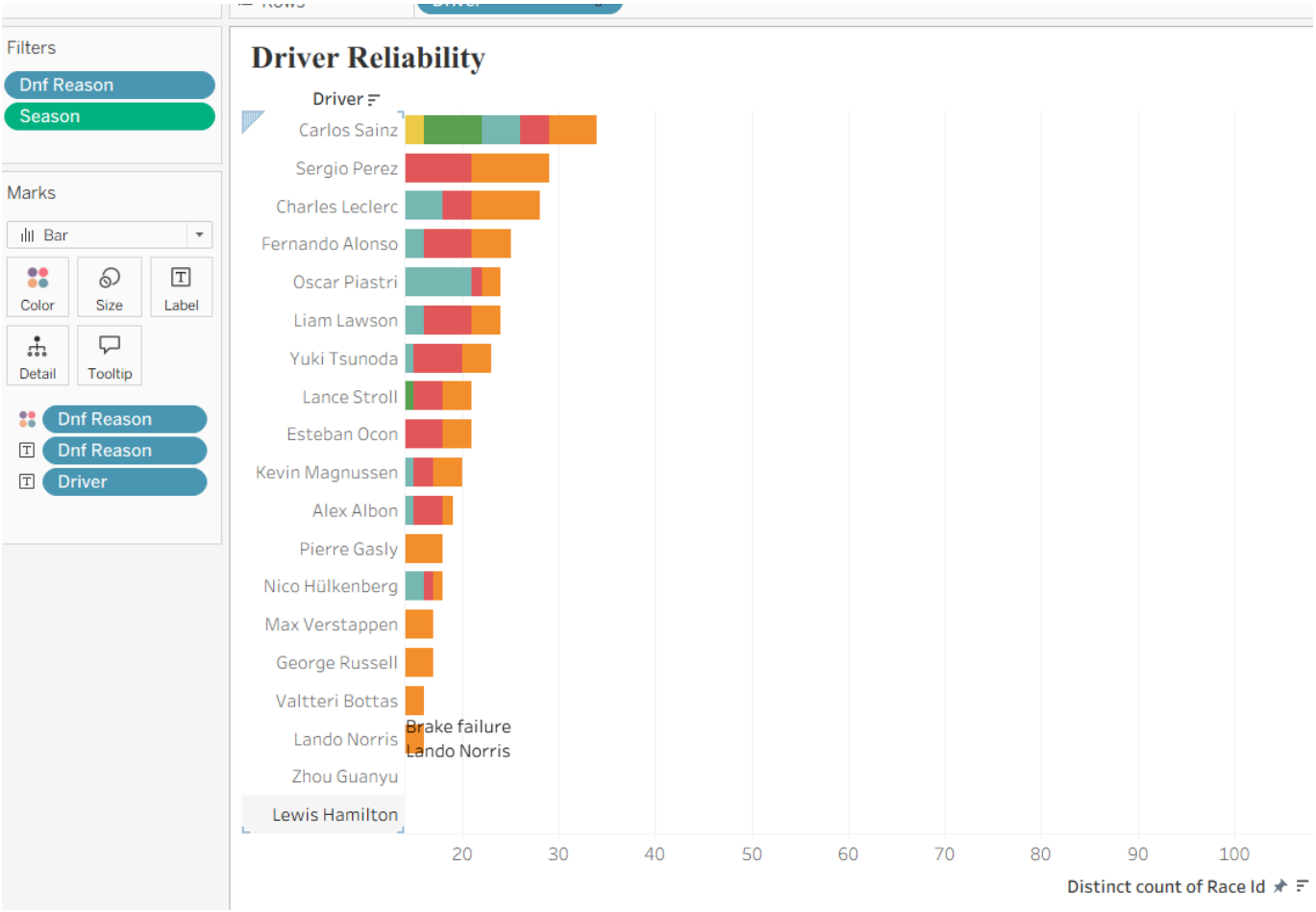


Figure 6. Driver Reliability and DNF Reasons — cropped from the supplied dashboard screenshot; taskbar excluded.

What it shows: The dashboard includes a driver reliability view and filters for retirement reasons such as mechanical, gearbox, hydraulics and engine-related issues. This allows investigation of reliability patterns by driver and selected reason.

Interpretation note: The driver names and marks are small in the screenshot; exact counts should be read directly in Tableau with filters selected.

6. Host Country Map

Business question: Where are the races in the dataset hosted geographically?

Visualization: Map showing host-country or race-location distribution.

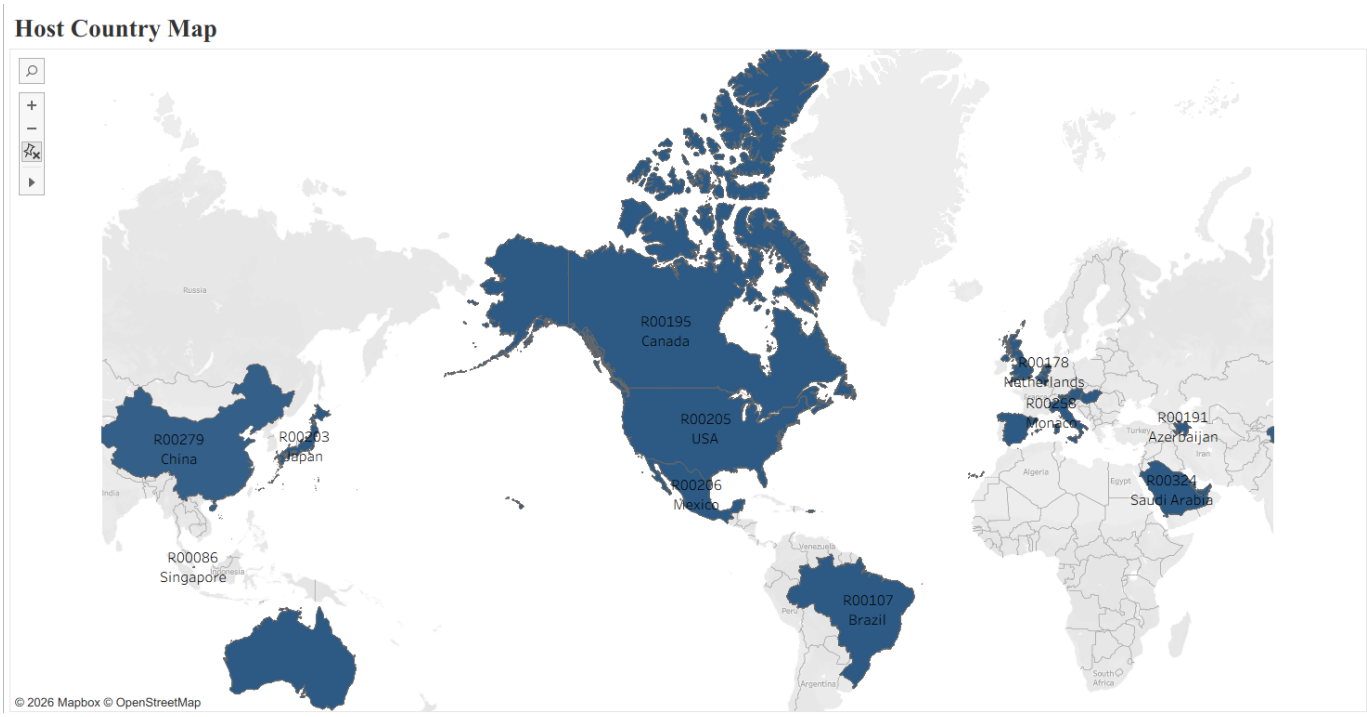


Figure 7. Host Country Map — cropped from the supplied dashboard screenshot; taskbar excluded.

What it shows: The map provides geographic context for race locations and helps identify the international spread of represented events. It is a map of host locations, not driver nationalities.

7. Completion Rate Trend

Business question: How does race completion rate vary across seasons?

Visualization: Completion-rate trend view with season on the horizontal axis.

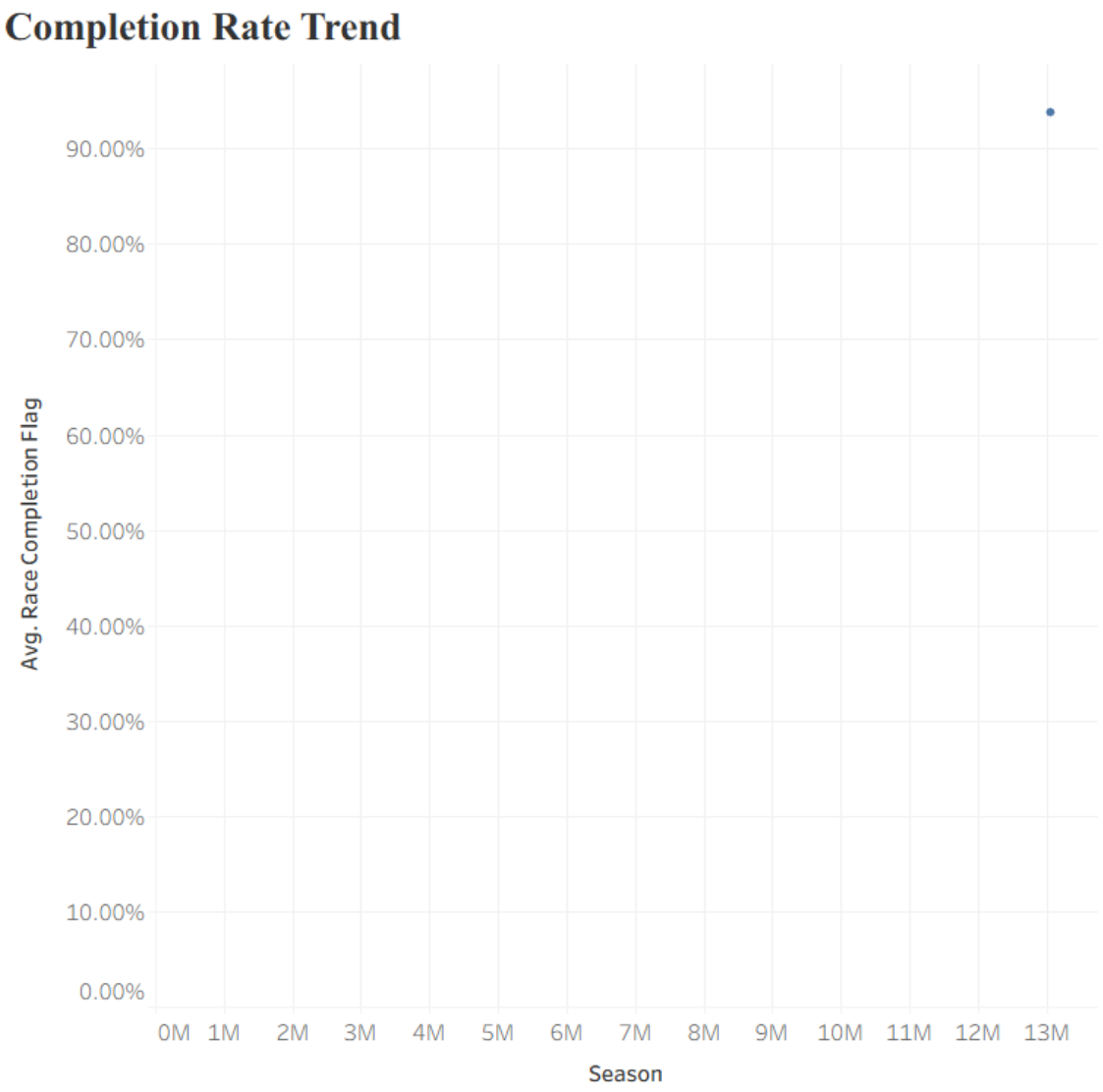


Figure 8. Completion Rate Trend — cropped from the supplied dashboard screenshot; taskbar excluded.

What it shows: The dashboard includes a completion-rate view intended to support season-by-season reliability analysis.

Interpretation note: The screenshot's axis/aggregation appears unclear and the plotted marks do not form an interpretable season-by-season trend. Verify the measure, season field and aggregation in Tableau before making a trend claim.

8. Season-wise Team Championship Performance

Business question: How has team performance varied across seasons?

Visualization: Stacked bar chart showing team points by season.

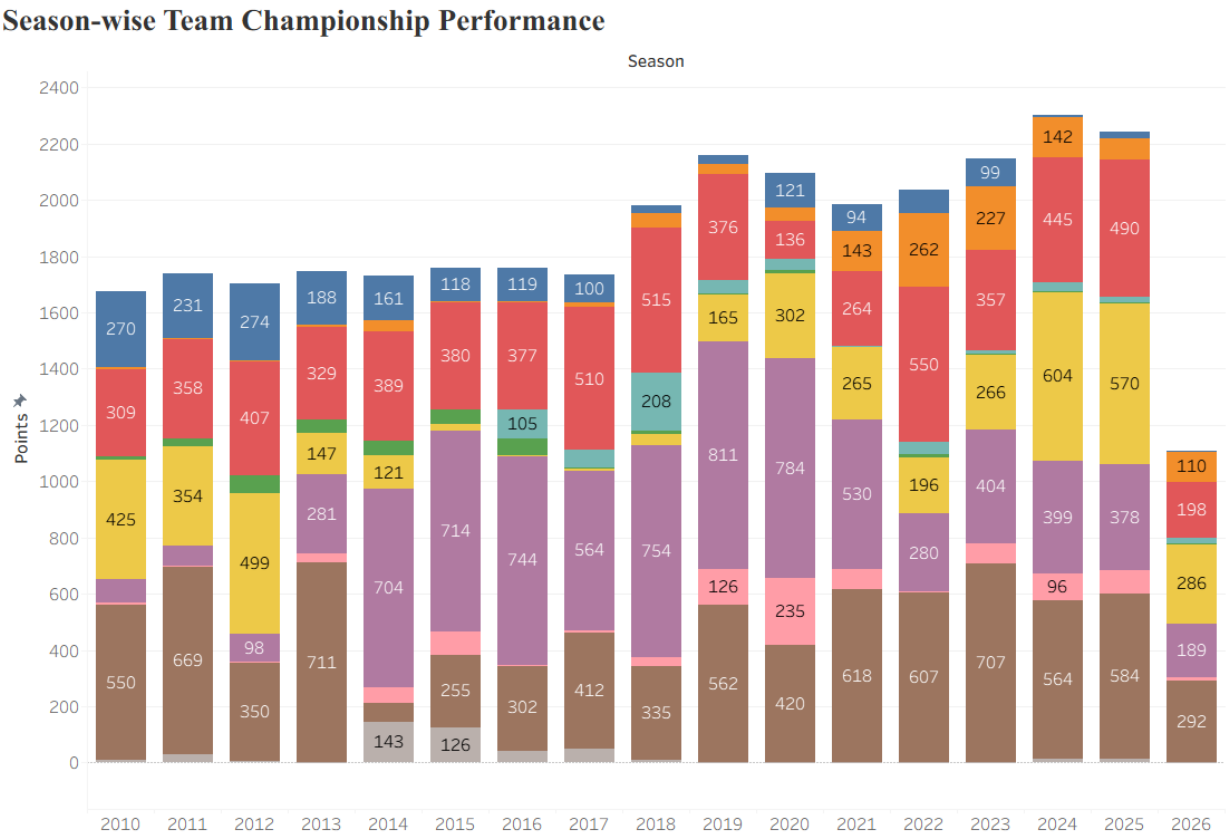


Figure 9. Season-wise Team Championship Performance — cropped from the supplied dashboard screenshot; taskbar excluded.

What it shows: The stacked bars compare the displayed team-points contribution across seasons and allow users to inspect changes in team performance over time.

9. Tyre Usage by Season

Business question: How does tyre usage change across seasons?

Visualization: Season-based tyre-usage view.

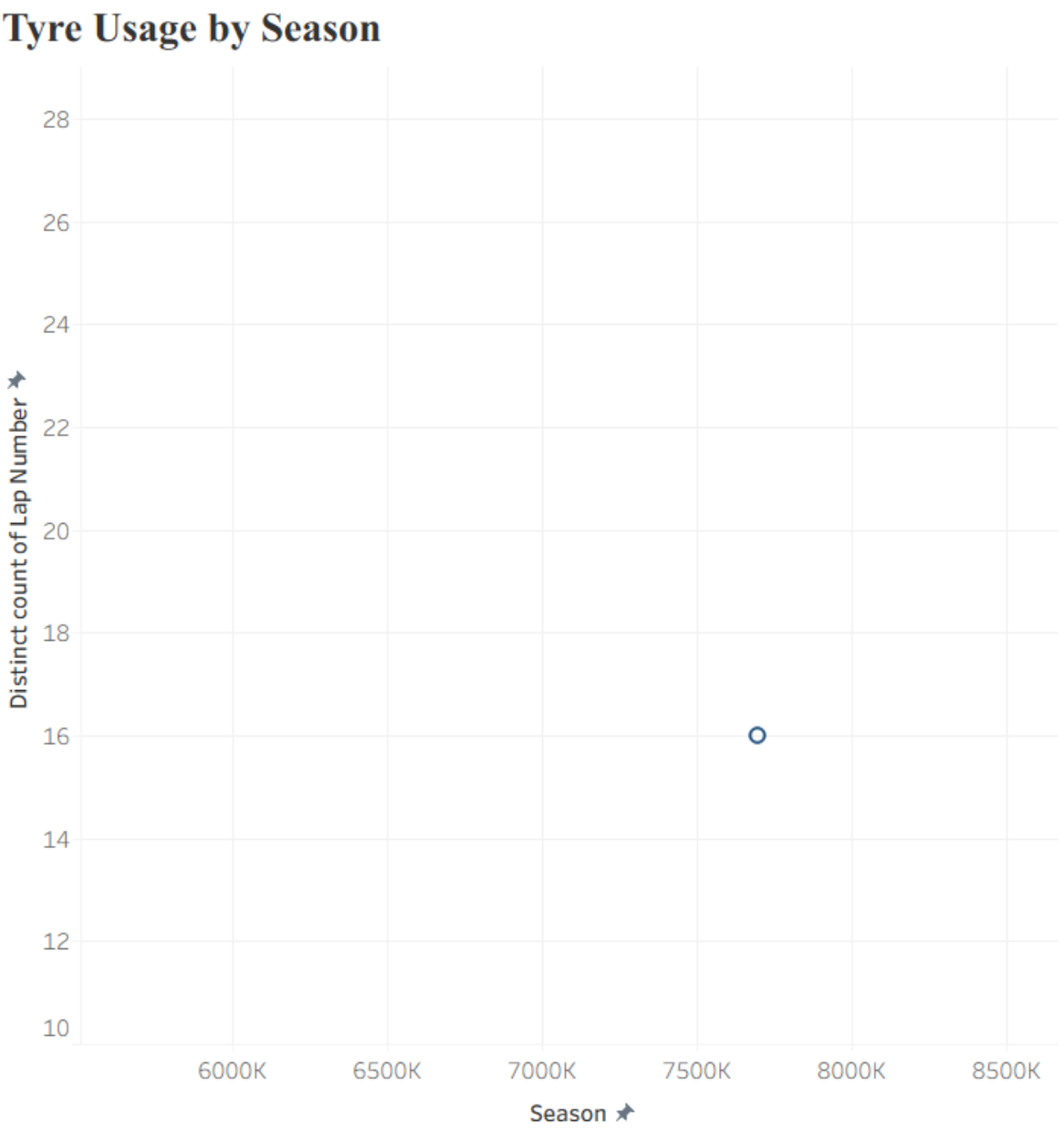


Figure 10. Tyre Usage by Season — cropped from the supplied dashboard screenshot; taskbar excluded.

What it shows: The chart is intended to support a comparison of tyre usage across seasons.

Interpretation note: The screenshot shows an unclear axis/aggregation and very few visible marks, so it does not currently support a reliable seasonal tyre-usage conclusion. Verify dimensions, measure and aggregation in Tableau.

4. Dashboard Design

Layout and visual hierarchy

The dashboard places summary KPIs along the top, the main performance charts across the central area, and filter controls on the right. This layout provides an overview first and then lets viewers explore specific subsets.

Visual encodings

Bar charts support comparison of drivers and teams; a pie chart summarizes tyre-compound shares; a bubble chart highlights podium-achievement differences; a map supplies geographic context; and a stacked bar chart compares team performance by season.

Interactive controls

Visible controls include Season, Team, Tyre Compound, DNF Reason, Race ID count and Tyre Age Laps. Users should apply filters deliberately and note that changing filters can change rankings and proportions.

Major outcomes

- Provides a single place to explore driver, team, tyre, podium, reliability and location information.
- Shows high-level counts for race records, seasons and circuits.
- Supports interactive exploration using season, team, tyre and DNF-related filters.
- Offers complementary views: rankings, proportions, time-based comparisons, a map and achievement bubbles.
- Flags for validation: the completion-rate and tyre-usage-by-season views should be checked for field aggregation and axis configuration before they are used for firm conclusions.

Dashboard public link

https://public.tableau.com/app/profile/gouri.sharma8417/viz/f1tableauproject/F1RACEINTELLIGENCEDASHBOARD

5. Tableau Story Design

The Tableau Story presents the analysis as a guided sequence rather than a single static view. A useful narrative order is: (1) dataset overview, (2) driver leaderboard, (3) team comparison, (4) tyre-compound distribution, (5) podium achievements, (6) reliability and DNF reasons, (7) host-country map, and (8) season-wise team performance.

Storytelling observations supported by the screenshot

- The dashboard's displayed KPI row reports 6,464 race records, 17 seasons analysed and 24 circuits represented.
- Soft, Medium and Hard tyres have similar displayed shares, while Intermediate tyres account for a smaller share.
- Red Bull Racing is the leading team in the visible team-points ranking, followed by Mercedes and Ferrari.
- The host-country map adds geographic context to the dataset.
- The season-wise team championship chart enables cross-season comparisons, while the completion-rate and tyre-usage-by-season views require validation before detailed conclusions are stated.

Published Tableau Story

https://public.tableau.com/views/f1tableauproject/Formula1RaceIntelligence

6. Web Integration and Project Repository

A Flask-based website provides access to the project and links to its Tableau dashboard/story. The project code is hosted in GitHub and the deployed website is available through Render.

- **Live website:** https://f1-race-intelligence-mhgm.onrender.com/
- **GitHub repository:** https://github.com/gourisanjaysharma-source/F1-Race-Intelligence

7. Data platform applicability

The supplied project-development pack also contains Snowflake-specific templates. The implementation evidence available for this project shows Tableau Public, a Flask website and a GitHub repository; it does not establish that Snowflake was used. Therefore, Snowflake SQL, data-preparation and Snowflake dashboard sections have not been fabricated. If the course requires Snowflake evidence, actual Snowflake tables, SQL scripts and query outputs must be supplied separately.

8. Final review checklist

- Confirm the published dashboard and story open correctly from the links above.
- Verify the aggregation and axes in Completion Rate Trend and Tyre Usage by Season.
- Check and address the null mark shown in Podium Achievements if it affects the intended interpretation.
- Apply representative filter combinations and confirm the KPI and chart values respond as expected.
- Add the actual Team ID if required by the submission portal; it was not present in the supplied material.